

Linux Network Security

Source: "Linux Security HOWTO"
available at
www.tldp.org/HOWTO/Security-HOWTO/network-security.html

Network security is becoming more and more important as people spend more and more time connected. Compromising network security is often much easier than compromising physical or local security, and is much more common.

There are a number of good tools to assist with network security, and more and more of them are shipping with Linux distributions.

Packet Sniffers

One of the most common ways intruders gain access to more systems on your network is by employing a packet sniffer on a already compromised host. This "sniffer" just listens on the Ethernet port for things like passwd and login and su in the packet stream and then logs the traffic after that. This way, attackers gain passwords for systems they are not even attempting to break into. Clear-text passwords are very vulnerable to this attack.

Example: Host A has been compromised. Attacker installs a sniffer. Sniffer picks up admin logging into Host B from Host C. It gets the admin's personal password as they login to B. Then, the admin does a su to fix a problem. They now have the root password for Host B. Later the admin lets someone telnet from his account to Host Z on another site. Now the attacker has a password/login on Host Z.

In this day and age, the attacker doesn't even need to compromise a system to do this: they could also bring a laptop or pc into a building and tap into your net.

Using ssh or other encrypted password methods thwarts this attack. Things like APOP for POP accounts also prevents